

The Critical 2D Stochastic Heat Flow

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Nikos Zygouras (Warwick)

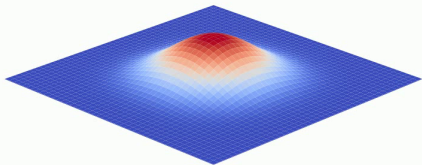
Rongfeng Sun (NUS)

Heat Equation

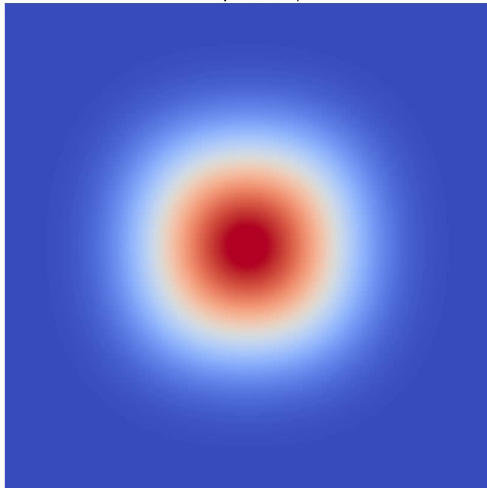
$$\partial_t u(t, x) = \Delta_x u(t, x)$$

(video by Q. Berger)

Density after n=14448 steps



Heat map after n=14448 step



In a nutshell

Stochastic Heat Equation

ill-defined in space dim. $d \geq 2$

$$\partial_t u(t, x) = \underbrace{\Delta_x u(t, x)}_{\sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} u(t, x)} + \underbrace{\beta \xi(t, x)}_{\text{"space-time white noise"} \atop \text{Gaussian cov. } \delta(t - t') \delta(x - x')} \cdot u(t, x)$$

We construct a “non-trivial solution” in space dimension $d = 2$

Critical 2D Stochastic Heat Flow (SHF)

Singular SPDEs



Statistical Mechanics

Outline

1. 2D Stochastic Heat Equation
2. Construction of SHF
3. Properties of SHF
4. SHF and Directed Polymers

Why is it interesting?

$$\partial_t u(t, x) = \Delta_x u(t, x) + \beta \xi(t, x) \cdot u(t, x)$$

► Fundamental PDE + universal random potential

► KPZ equation

[Kardar–Parisi–Zhang 86]

$$\partial_t h(t, x) = \Delta_x h(t, x) + |\nabla_x h(t, x)|^2 + \beta \xi(t, x)$$

Cole–Hopf: $h(t, x) = \log u(t, x)$

[Corwin 12] [Quastel–Spohn 15]

Why is it difficult?

$$\partial_t u(t, x) = \Delta_x u(t, x) + \beta \xi(t, x) \cdot u(t, x)$$

$$\xi(t, x) \text{ is a distribution} \rightsquigarrow u(t, x) \text{ expected } \begin{cases} \text{non-smooth function} & d = 1 \\ \text{genuine distribution} & d \geq 2 \end{cases}$$

(additive case)

Ill-defined product $\xi(t, x) \cdot u(t, x) \rightsquigarrow$ Singular SPDEs

How to make sense of the equation?

$$u(0, y) = \psi_0(y)$$

Stochastic integral

Duhamel's principle: integral formulation

$g_t(x)$ = heat kernel

$$u(t, x) = \int_{\mathbb{R}^d} g_t(x-y) \underbrace{\psi_0(y)}_{\text{initial datum}} dz + \iint_{(0,t) \times \mathbb{R}^d} g_{t-s}(x-y) u(s, y) \beta \xi(s, y) ds dy$$

Works for $d = 1$

- ▶ stochastic integral [Ito/Walsh] [Da Prato–Zabczyk]
- ▶ well-posedness, positivity [Chen–Dalang 15] [Müller 91]

Breaks down for $d \geq 2$

$g_t(x) \notin L^2(dt, dx)$

Pathwise stochastic analysis

Revolution in 2010s: robust solution theories for **subcritical singular SPDEs**

- ▶ Regularity Structures [Hairer]
- ▶ Paracontrolled Distributions [Gubinelli–Imkeller–Perkowski]
- ▶ Energy Solutions [Gonçalves–Jara]
- ▶ Renormalization Group Approach [Kupiainen, Duch]

Stochastic Heat Equation and KPZ are **subcritical** only for $d = 1$

These approaches **do not apply** for $d \geq 2$

The role of dimension

$$\partial_t u(t, x) = \Delta_x u(t, x) + \beta \xi(t, x) u(t, x)$$

Space-time blow-up $\tilde{u}(t, x) := u(\delta^2 t, \delta x)$

$$\partial_t \tilde{u}(t, x) = \Delta_x \tilde{u}(t, x) + \delta^{\frac{2-d}{2}} \beta \tilde{\xi}(t, x) \tilde{u}(t, x)$$

As $\delta \downarrow 0$ the noise strength $\begin{cases} \text{vanishes} & d < 2 & \text{sub-critical} \\ \text{unchanged} & d = 2 & \text{critical} \\ \text{diverges} & d > 2 & \text{super-critical} \end{cases}$

We will henceforth focus on the **critical dimension** $d = 2$

What can we do?

Regularized noise $\xi_\varepsilon(t, x)$ on scale $\varepsilon > 0$ (discretization, mollification, ...)

Well-defined equation
$$\begin{cases} \partial_t u_\varepsilon(t, x) = \Delta_x u_\varepsilon(t, x) + \beta \xi_\varepsilon(t, x) u_\varepsilon(t, x) \\ u_\varepsilon(0, x) = \psi_0(x) \geq 0 \in C_b \quad (\text{e.g. } \psi_0 \equiv 1) \end{cases}$$

Fix $\xi_\varepsilon(t, x) \xrightarrow{\varepsilon \downarrow 0} \xi(t, x)$ Does the solution u_ε converge?

Nontrivial answer: depends subtly on the noise strength β

Which notion of convergence?

Space-average $u_\varepsilon(t, \varphi) := \int_{\mathbb{R}^2} \varphi(x) u_\varepsilon(t, x) \, dx \xrightarrow[\varepsilon \downarrow 0]{d} \mathcal{U}(t, \varphi) \quad \varphi \in C_c$

$\Longleftrightarrow u_\varepsilon(t, x) \, dx \xrightarrow[\varepsilon \downarrow 0]{d} \mathcal{U}(t, dx)$ vague convergence
of random measures

Theorem

[Berger–C.–Turchi 25]

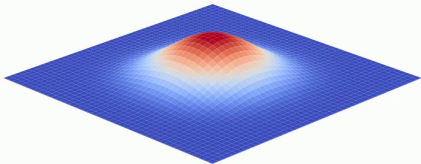
Any **fixed** noise strength $\beta \geq 0$: **trivial limit** $\mathcal{U} =$ *non random*

$$u_\varepsilon(t, \cdot) \xrightarrow[\varepsilon \downarrow 0]{} \begin{cases} \text{sol. of } \textbf{deterministic} \text{ heat equation} & \text{if } \beta = 0 \\ \mathbf{0} & \text{if } \beta > 0 \end{cases}$$

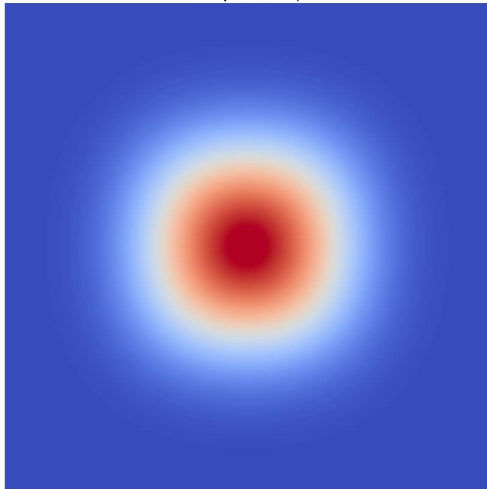
Discretization: $\beta = 0$

(by Q. Berger)

Density after n=14448 steps



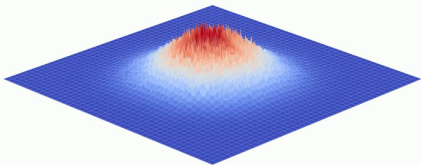
Heat map after n=14448 step



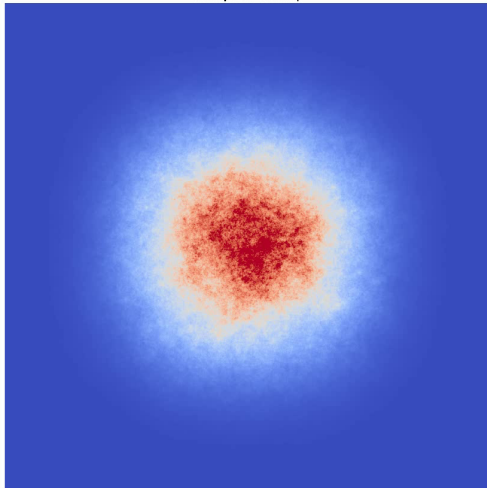
Discretization: β very small

(by Q. Berger)

Density after $n=14448$ steps

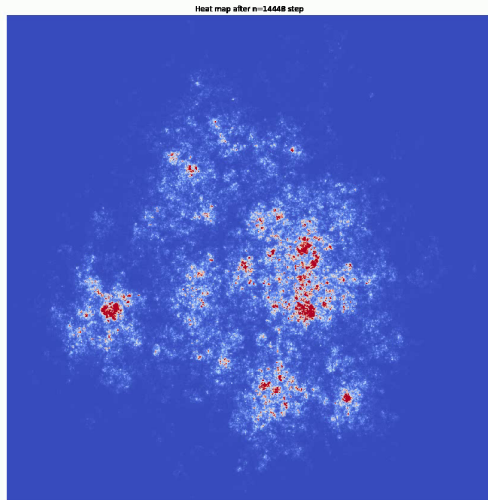
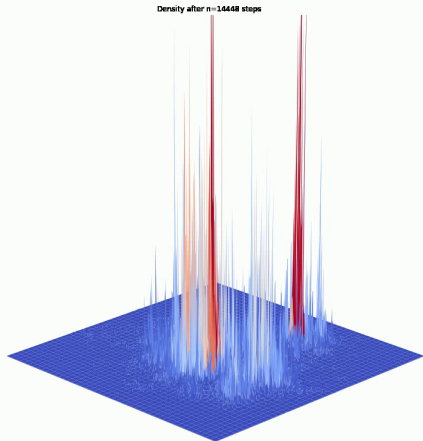


Heat map after $n=14448$ step



Discretization: β “critical”

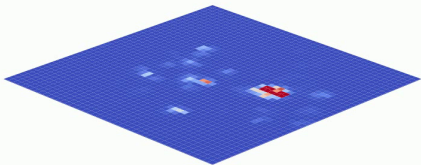
(by Q. Berger)



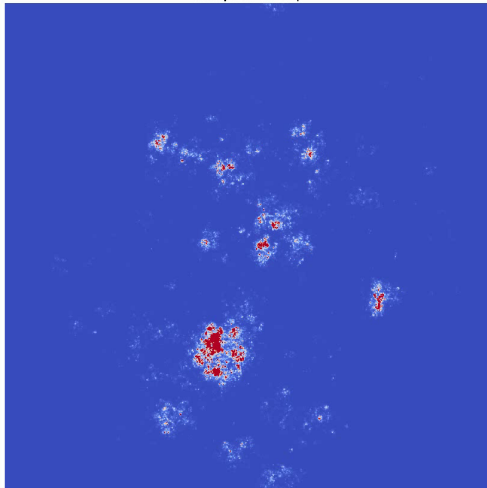
Discretization: β “supercritical”

(by Q. Berger)

Density after $n=14448$ steps



Heat map after $n=14448$ step



The need for renormalization

Critical scaling

($\rightsquigarrow$ phase transition)

$$\beta = \beta_\varepsilon = \frac{\hat{\beta}_{\text{crit}}}{\sqrt{|\log \varepsilon|}} \left(1 + \frac{\vartheta}{|\log \varepsilon|} \right)$$

our discretization

$$\hat{\beta}_{\text{crit}} = \sqrt{\pi}$$

Noise strength vanishes as $\varepsilon \downarrow 0$, yet **the effect of noise persists**

$$\blacktriangleright \text{Var}[u_\varepsilon(t, \varphi)] \xrightarrow{\varepsilon \downarrow 0} K_t^\vartheta(\varphi, \varphi) > 0$$

[Bertini–Cancrini 98] [C.S.Z. 19]

$\blacktriangleright$ All moments converge

[C.S.Z. 19] [Gu–Quastel–Tsai 21]

$$\mathbb{E}[u_\varepsilon(t, \varphi)^q] \geq \exp(\exp(cq))$$

[Ganguly–Nam 25]

Main result

Theorem

[C.S.Z. 23]

Discretize the **Stochastic Heat Equation** on the lattice $\varepsilon \mathbb{N} \times \sqrt{\varepsilon} \mathbb{Z}^2$

Rescale
$$\beta_\varepsilon = \frac{\sqrt{\pi}}{\sqrt{|\log \varepsilon|}} \left(1 + \frac{\vartheta}{|\log \varepsilon|} \right) \quad \vartheta \in \mathbb{R}$$

The solution u_ε converges to a **unique** and **non-trivial limit** SHF^ϑ

$$u_\varepsilon = \left(u_\varepsilon(t, x) dx \right)_{t \geq 0} \xrightarrow[\varepsilon \downarrow 0]{d} \text{SHF}^\vartheta = \left(\text{SHF}^\vartheta(t, dx) \right)_{t \geq 0}$$

Measure-valued process on $\mathbb{R}^2$

Critical 2D **Stochastic Heat Flow**

Techniques and Ideas

Subsequential limits exist by **tightness**

via moments bounds

Challenge is **uniqueness**

lacking characterization of the limit

What we proved

[C.S.Z. 23]

$(u_\varepsilon)_{\varepsilon>0}$ is **Cauchy**

$u_\varepsilon \approx u_{\varepsilon'}$

for small $\varepsilon, \varepsilon' > 0$

► Coarse-graining

approximate **self-similarity**

► Lindeberg principle

robustness to microscopic details

These ideas played an important role in subsequent developments

SHF and Stochastic Heat Equation

SHF = limit of solutions of **renormalized** 2D Stochastic Heat Equation

$$\text{SHF}^\vartheta(t, dx) \quad \text{SHF}^\vartheta(0, \psi_0; t, dx) = \int_{\mathbb{R}^2} \psi_0(y) \text{SHF}^\vartheta(0, dy; t, dx)$$

$\rightsquigarrow$ 2 space-time parameters $\left(\text{SHF}^\vartheta(s, dy; t, dx) \right)_{0 \leq s \leq t < \infty}$

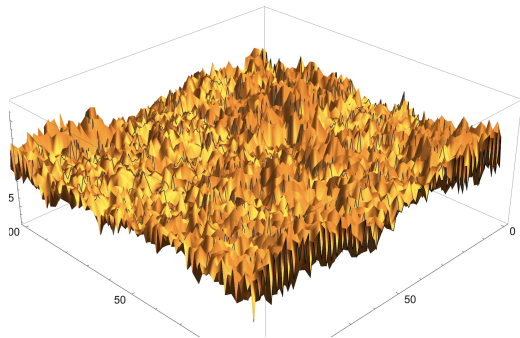
Flow property / Chapman–Kolmogorov: for $s < t < u$

[Clark–Mian 26]

$$\text{SHF}^\vartheta(s, dy; u, dz) = \int_{x \in \mathbb{R}^2} \underbrace{\text{SHF}^\vartheta(s, dy; t, dx) \text{SHF}^\vartheta(t, dx; u, dz)}_{\text{non-trivial "product" of measures}}$$

Some **not-so-randomly picked** realizations

[M. Mucciconi, N. Zygouras]



$$\text{KPZ} \approx \log u_\varepsilon(t, x)$$



Alps, Italy

(by Mg-k)

Universality of SHF

We constructed SHF^ϑ via **discretization** of the Stochastic Heat Equation

Universality

Other regularizations should also lead to SHF^ϑ

Theorem

[Tsay 24]

Stochastic Heat Equation with **mollified** noise $\xi_\varepsilon = \xi * \rho_\varepsilon$

Rescale
$$\beta_\varepsilon = \frac{\sqrt{2\pi}}{\sqrt{|\log \varepsilon|}} \left(1 + \frac{\vartheta + c}{|\log \varepsilon|} \right) \quad \vartheta \in \mathbb{R}$$

The solution u_ε converges to SHF^ϑ as $\varepsilon \downarrow 0$

$\rightsquigarrow$ Far reaching universality

[Drillick, Hou, Parekh 26]

Characterization of SHF

Theorem

[Tsay 24]

Let $\mathcal{Z} = (\mathcal{Z}_{s,t}(dx, dy))_{s \leq t}$ be a stochastic process on $\mathcal{M}_+(\mathbb{R}^2 \times \mathbb{R}^2)$ satisfying

- ▶ **continuity** of $(s, t) \mapsto \mathcal{Z}_{s,t}$
- ▶ **independence** of $\mathcal{Z}_{s_1, t_1}, \mathcal{Z}_{s_2, t_2}, \dots, \mathcal{Z}_{s_k, t_k}$ $\forall s_1 < t_1 < \dots < s_k < t_k$
- ▶ **flow property** (Chapman-Kolmogorov) $\mathcal{Z}_{s,u} = \mathcal{Z}_{s,t} \bullet \mathcal{Z}_{t,u}$ $\forall s < t < u$
- ▶ **same moments** $\mathbb{E}[\prod_{i=1}^n \mathcal{Z}_{s_i, t_i}(\varphi_i, \psi_i)]$ as SHF^{ϑ} up to $n = 4$

Then

$$\mathcal{Z} \stackrel{d}{=} \text{SHF}^{\vartheta}$$

SHF and noise

Stochastic Heat Equation with **regularized noise** (discretized/mollified)

$$\partial_t u_\varepsilon(t, x) = \Delta_x u_\varepsilon(t, x) + \beta_\varepsilon u_\varepsilon(t, x) \xi_\varepsilon(t, x)$$

As $\varepsilon \downarrow 0$ $u_\varepsilon \longrightarrow \text{SHF}^\vartheta$ $\xi_\varepsilon \longrightarrow \xi$ $\beta_\varepsilon = \beta_\varepsilon(\vartheta) \longrightarrow 0$

Theorem

[C.–Donadini 25] [Gu–Tsai 25]

$$(u_\varepsilon, \xi_\varepsilon) \xrightarrow[\varepsilon \downarrow 0]{d} (\text{SHF}^\vartheta, \xi) \quad \text{with } \text{SHF}^\vartheta \text{ and } \xi \text{ independent}$$

$\rightsquigarrow$ **No SPDE** driven by ξ for SHF^ϑ

► Noise sensitivity [C.–Donadini 25]

► Black noise [Gu–Tsai 25]

Basic Properties of SHF

► $\text{SHF}^{\vartheta}(t, dx) \neq \text{Gaussian}$ (positive random measure)

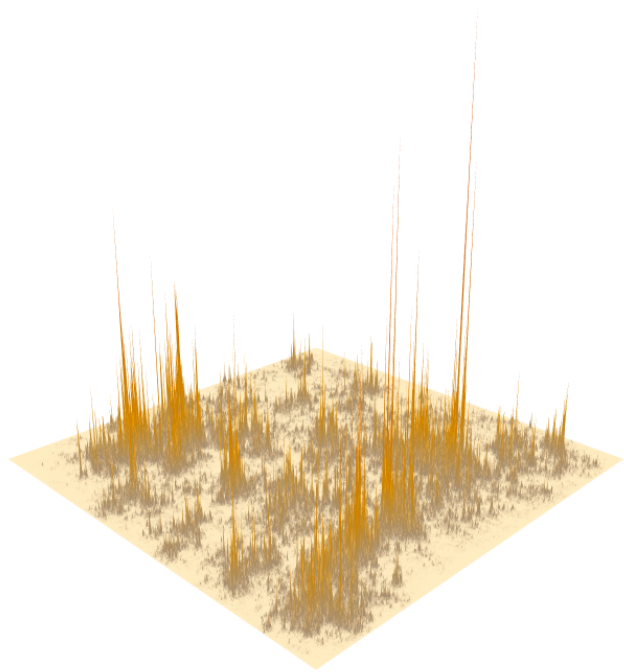
► $\text{SHF}^{\vartheta}(t, dx) \neq \text{GMC} \simeq \exp(\text{Gaussian})$ [C.S.Z. 23]

But **conditional GMC structure** on path space [Clark–Tsai 25]

► **Moment formulas** [C.S.Z. 19] [Gu–Quastel–Tsai 21] [Ganguly–Nam 25]

$\rightsquigarrow$ [Cosco, Nakajima, Zeitouni] [Lygkonis, Zygouras] [Liu, Zygouras] ...

► **Scaling covariance**
$$\frac{\text{SHF}^{\vartheta}(\alpha t, d(\sqrt{\alpha}x))}{\alpha} \stackrel{d}{=} \text{SHF}^{\vartheta + \log \alpha}(t, dx)$$



Singularity and Regularity

- ▶ $\text{SHF}^{\vartheta}(t, dx)$ **singular** w.r.t. Lebesgue [C.S.Z. 25+]
“not a function”
- ▶ $\text{SHF}^{\vartheta}(t, dx) \in \mathcal{C}^{-\kappa}$ for any $\kappa > 0$ $\rightsquigarrow$ non atomic
“barely not a function”
- ▶ **Small ball** $\text{SHF}^{\vartheta}(1, g_{\varepsilon^2}) = |\log \varepsilon|^{-1/2+o(1)}$ [Gu–Tsai 26]
+ Gaussian fluctuations
- ▶ **Strict positivity** $\text{SHF}^{\vartheta}(t, B) > 0 \quad \forall \text{ ball } B$ [Clark–Tsai 25]
[Nakashima 25]

More

► Strong disorder $\text{SHF}^\vartheta(t, B) \xrightarrow[\vartheta \rightarrow \infty]{d} 0 \quad \forall \text{ ball } B$ [Clark–Tsai 25]

with sharp quantitative rates [Berger.–C.–Turchi 25]

► Weak disorder $\text{SHF}^\vartheta(t, dx) \xrightarrow[\vartheta \rightarrow -\infty]{d} dx$ for $\psi_0 \equiv 1$

+ Gaussian fluctuations (Edwards–Wilkinson) [C.–Cottini–Rossi 25]

► Martingale problem [Nakashima 25] [Chen 25]

► (Multi-)Fractal properties [Ganguly–Nam 26] [Ganguly–Xie–Zygouras 26+]

...

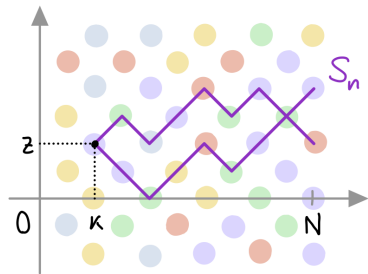
Directed Polymer in Random Environment

► Simple random walk on $\mathbb{Z}^2$ $S = (S_n)_{n \geq 0}$

► Independent Gaussians $\omega(n, x) \sim \mathcal{N}(0, 1)$

► Interaction $H_N(S, \omega) := \sum_{n \leq N} \omega(n, S_n)$

Polymer measure $P_N(S) = \frac{e^{\beta H_N(S, \omega)}}{Z_N} P^{\text{SRW}}(S)$



Partition Functions

$$Z_N(k, z) = \mathbb{E}_{(k, z)} \left[e^{\beta H_N(S, \omega)} \right] / \mathbb{E}[\dots]$$

Stochastic Heat Equation and Feynman–Kac

Partition functions solve discretized Stochastic Heat Equation (Duhamel)

$$-\partial_k Z_N(k, z) = \Delta_z Z_N(k, z) + (e^{\beta \omega(k, z)} - 1) Z_N(k, z)$$

Discretized Feynman–Kac

$$Z_N = \mathbb{E} \left[e^{\beta H_N(S, \omega)} \right] \approx \mathbb{E} \left[e^{\beta \int \xi(s, B_s) ds} \right]$$

Diffusive rescaling by factor $N \rightsquigarrow$ Discretization on scale $\varepsilon = \frac{1}{\sqrt{N}}$

$$Z(N(1-t), \sqrt{N}x) = u_\varepsilon(t, x)$$

Conclusion

Probabilistic handle on Stochastic Heat Equation via Directed Polymers

$\rightsquigarrow$ KPZ universality in $d = 1$ [Seppäläinen 12] [Alberts–Khanin–Quastel 14]

Statistical Mechanics $\longleftrightarrow$ Singular SPDEs

Disorder (ir)relevance $\longleftrightarrow$ super/subcriticality

This goes much beyond the SHF

A broader picture

Extensions of Stochastic Heat Equation

- ▶ Nonlinear [Dunlap–Gu 22] [Dunlap–Graham 23]
- ▶ Lévy noise [Berger–Chong–Lacoin 23]
- ▶ Correlated Gaussian noise [Müller–Tribe 04] [Dunlap–Hairer–Li 25]
[Cosco–Cottini–Donadini 25]

Critical SPDEs

- ▶ Anisotropic KPZ equation [Cannizzaro–Erhard–Toninelli 23]
- ▶ Stochastic Burgers equation [Cannizzaro–Gubinelli/Moulard–Toninelli 24 25]
- ▶ Diffusion in the curl of GFF [Chatzigeorgiou–Morfe–Otto–Wang 25]
[Armstrong–Bou Rabee–Kuusi 24]
- ▶ 4D Parabolic Anderson Model [Gabriel–Rosati 26] [Deng–Shen 26]

Thanks